Algorithm Frontier

A curated catalogue of exact algorithms, tradeoffs, and conditional frontiers

Source-linked research catalogue

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Abstract

This catalogue collects exact algorithms, approximation schemes, dynamic graph data structures, and lower-bound records in a common notation. The entries are organized by problem family. Each statement names its input model, output contract, controlling parameters, and the resource bound that is actually claimed; incomparable time–space or update–stretch points remain separate. The repository cards and source maps linked below are the authoritative detailed records. This document is a compact mathematical index, rather than a proof of any result.

1 Scope and notation

The input-size symbol H denotes the complete binary encoding together with the polynomial work charged to the stated oracle or primitive. Polynomial factors in H are suppressed only when the displayed exponential term is the point of the record. The following symbols recur throughout the catalogue:

Symbol	Meaning	Symbol	Meaning
n, m	vertices and edge records	L	binary length of a subset-sum instance
T, Q	target budget and support-pattern count	B, B_0	maximum and root quota caps
Δ, p	nonroot incidence and shared-family counts	U, h	logarithmic history and hierarchy scales
r, q, c	selected rank, item count, and rejected count	ε	rational approximation accuracy

A bound on time, writable space, output length, or update work is labelled explicitly. A conditional row includes the backend or oracle contract in its statement. Lower-bound scope is recorded separately from an upper bound.

2 Subset Sum

Both records solve signed binary Subset Sum exactly and return an indexed witness on YES instances. Repeated, zero, and negative weights are part of the input model; n counts weights and L counts the complete encoding.

Curated record (faster deterministic point) *For a target t, decide whether some subset of the n weights sums to t and return its original indices. The reviewed time envelope is*

$$\mathrm{poly}(L) + \mathrm{poly}(n, L)\, 2^{En}, \qquad E = \frac{149809813343531}{301754962128739} \approx 0.4964618056.$$

No separate space exponent is claimed. This point is therefore a time record, not a simultaneous time–space guarantee. See the faster-time card and its source map.

Curated record (deterministic low-space point) *The same exact decision and indexed-witness task has*

$$\textit{time} = \mathrm{poly}(L)\, 2^{n/2}, \qquad \textit{writable space} = \mathrm{poly}(L)\, 2^{E_S n},$$
$$E_S = \frac{18261635770550353}{10^{17}} = 0.18261635770550353.$$

Read-only input storage is excluded from the writable-space row. The exact space exponent supersedes the weaker $137/750$ *presentation. See the low-space card.*

3 Numeric targets and quota systems

These families separate global target budgets from ordered or sparse quota geometry. Binary caps are parameters; they are not expanded into unit-sized copies.

3.1 Target and support-sensitive polymatroids

The numeric-target record gives exact target feasibility and scalar optimization, with approximation and support-sensitive refinements. In the arbitrary-target row the envelope is exponential in T; fixed row dimension d yields a sharper stretched-exponential row, and Q distinct eligible-column support patterns replace an arity restriction in the support-sensitive row. The common forms include

$$2^{O((T+1)\log(T+2))}\,\mathrm{poly}(H) \quad \text{(arbitrary target)}, \qquad 2^{O(T\log(2+Q/T))}\,\mathrm{poly}(H) \quad \text{(support-sensitive row)}.$$

Each approximation statement carries its own ε dependence; the source does not identify approximation with exact decision. The complete parameter contract and the two rank-oracle lower-bound scopes are in the numeric-target card.

3.2 Weighted prefixes, laminar caps, and sparse quotas

For the ordered quota family, B is the largest target/root cap, B_0 is a distinguished root cap, $\Delta \geq 1$ bounds positive nonroot incidences per column, and p counts shared weight families. The reviewed end-to-end rows include

$$\begin{aligned} \text{one order} &: 2^{O(B)}\,\mathrm{poly}(H), & \text{two orders} &: 2^{O(B\log(B+2))}\,\mathrm{poly}(H),\\ p \text{ families} &: 2^{O(pB\log(p+2))}\,\mathrm{poly}(H), & \text{laminar root} &: 2^{O(B^2\log(B+1))}\,\mathrm{poly}(H). \end{aligned}$$

Sparse mixed quotas retain a separate support-sensitive bound, with the incidence and root-cap dependence written in the card rather than hidden in the polynomial factor. Ground-kernel outputs (packing, coverable-matroid, transversal, and matching kernels) are separate rows with their own parameter and output contracts. See the weighted-prefix card.

4 Dynamic graph algorithms

The dynamic records preserve the distinction between amortized update work, stretch, recourse, and query cost. Here N is a polynomial history cap, $U = \log N + O(1)$, and $h = \log(U + O(1))$.

4.1 All-pairs distances and spanners

For fully dynamic APSP, the equalized point has amortized update work and metric stretch of the form

$$\exp\bigl(O(U^{4/5}h^{1/5})\bigr), \qquad \text{query time} = \mathrm{poly}(U).$$

For every fixed positive constant C, the second point trades these terms as

$$\text{update} = \exp\bigl(O(U^{3/4}h^{(C+1)/4})\bigr), \qquad \text{stretch} = \exp\bigl(O(U/h^{C})\bigr).$$

These are incomparable upper-bound points; neither is asserted to be optimal. The APSP card defines n, m and the positive polynomial-range edge-length model.

Dynamic spanners retain two constructions because their recourse contracts differ. With m_0 initial edges and I, D insertions and deletions, the tracked-terminal row uses an $\exp(O(U^{4/5}h^{1/5}))$ envelope, while the full-horizon row uses $\exp(O(U^{5/6}h^{1/6}))$. The latter bounds output recourse independently of the cumulative insertion count; the former tracks terminal changes throughout the declared history. See the spanner card.

4.2 Incremental shortest paths and connectivity

The incremental directed SSSP record is conditional on a frozen detector and retirement interface. It returns a simple path with original edge identifiers, and its total work is

$$O(n+m) + (n+m)\exp\left(O\left(\log E(N) + \sqrt{U\left(h + \log(1/\varepsilon)\right)}\right)\right),$$

where the candidate backend has $\log E(N) = O(U^{4/5}h^{1/5})$ and $0 < \varepsilon \le 1/2$. Incremental SCC, decremental strong-forest, and dynamic MSF cards keep their own update model and certificate assumptions; they are indexed from the family guide.

5 Matching and graph optimisation

Exact general-graph matching is a conditional composition. For m_s retained edge records after normalization, its backend-dependent time is

$$O(n+m) + (m_s+1)\exp\left(O\left((\log(m_s+2))^{4/5}(\log\log(m_s+4))^{1/5}\right)\right).$$

The output preserves original edge identifiers, including parallel lifetimes. The matching card states the raw-backend contract; no new unconditional matching lower bound is claimed.

Degree-constrained subgraphs transfer the same backend to f-factors, interval degree bounds, and cardinality optimization in an explicit loopless multigraph. Static and threshold min-cost flow use integral capacities, costs, and divergence with binary encoding; their raw-cycle lifetime interface yields

$$O((N+1)E(S)^4\,\mathrm{poly}(\log S)), \qquad E(S) = \exp\left(O((\log S)^{4/5}(\log\log(S+4))^{1/5})\right).$$

Full-flow output is charged separately. See the degree card and flow card.

6 Allocation, robust flow, and scheduling

The remaining records preserve scenario, selector, and binary-count parameters because these contracts change the theorem. Selected splittable flow chooses r of q commodities, rejects $c = q - r$, and returns a common rate at least $(1-\varepsilon)$ times optimum. Its selector base is raised to c:

$$\Gamma(1/\varepsilon)^c\ \mathrm{poly}(H),$$

with the selector, routing budgets, and rational-data model stated separately in the selected-flow card. Adaptive robust flow uses one common selected set and scenario-specific routings; the explicit scenario list is part of H and exact full-rate admission hardness is kept separate.

For arbitrary recipient polymatroids, the principal allocation ratio is

$$d = 4 - \frac{3}{r}, \qquad \text{value} \ge \frac{\mathrm{OPT}}{d}, \qquad \text{time} = 2^{O(r)}\,\mathrm{poly}(H),$$

with an optional $d + \eta$ refinement. Two-size scheduling has distinct corank and large-unit branches: factors $3/2$ and $5/3$, respectively, with times $2^{O(c)}\,\mathrm{poly}(H)$ and $2^{O(h)}\,\mathrm{poly}(H)$. These regimes are not merged. The allocation card and scheduling card give the exact oracle, decomposition, and lower-bound assumptions.

7 Reading the source records

Every displayed row links to a problem card whose proof map, source map, and review status are maintained with the result. The compact index is CONTENTS.md; family boundaries and the complete card list are in docs/FAMILIES.md. Finite enumeration and rational-certificate checks support selected proof components; they do not replace a proof of an asymptotic algorithm.

References

[1] Algorithm Frontier repository, `https://github.com/icekylinx/algorithm-frontier/`.

[2] Algorithm Frontier, "Problem families and result cards," docs/FAMILIES.md.

[3] Algorithm Frontier, "Complexity notation and parameter policy," docs/COMPLEXITY.md.

[4] Algorithm Frontier, verification scripts and finite certificates, tools/.

The repository cards are authoritative for hypotheses, proof details, and revisions.